CLASS 10 MATHEMATICS • CHAPTER 10

Circles

Exhaustive point-by-point presentation covering tangents, point of contact, radius-tangent perpendicularity, and properties of tangents from external points directly from the textbook.

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Line and Circle Positions in a Plane

What is a Circle?

- Collection of all points in a plane at a constant distance (radius) from a fixed point (centre).
- Learned terms: chord, segment, sector, arc, etc., in Class IX.

Three Possible Positions of a Line

- **Non-intersecting:** Line and circle have no common point.
- **Secant:** Line intersects the circle at exactly two points.
- **Tangent:** Line intersects the circle at exactly one point.

Section 10.1 Live Check

Q1. What are the three possible relationships between a straight line and a circle in a plane?

Q2. What is the distinguishing feature of a secant compared to a tangent?

Q3. How can the rope on both sides of a well pulley be modeled mathematically?

Defining Tangents & Point of Contact

Etymology & Core Definition

The word 'tangent' comes from Latin 'tangere' (to touch). A tangent touches a circle at **only one point** called the point of contact.

Secant Limit Case

A tangent is a special case of a secant when the two endpoints of its corresponding chord coincide at the point of contact.

Parallel Tangents Limit

There can be at most two parallel tangents to a circle for a given parallel secant direction.

Section 10.2 Live Check

Q1. How is a tangent related to a secant of a circle?

Q2. What do we call the unique common point where a tangent touches a circle?

Q3. How many tangents can be parallel to a given secant line on a circle?

Theorem 10.1 Statement & Proof

Theorem 10.1

The tangent at any point of a circle is **perpendicular to the radius** through the point of contact.

Proof Summary via Shortest Distance

- Take any point Q on tangent XY other than point of contact P.
- Q lies outside the circle, so OQ > OP (radius).
- Since this holds for every point on XY except P, OP is the **shortest distance** from O to XY.
- Therefore, OP ⊥ XY. (Normal is another name for this line).

Theorem 10.1 Live Check

Q1. State Theorem 10.1 regarding the angle between a tangent and the radius at the point of contact.

Q2. Why is a point Q on the tangent (other than the point of contact P) always outside the circle?

Q3. What alternative name is sometimes given to the line containing the radius through the point of contact?

Tangents from Internal, On-Circle, and External Points

Three Cases

- **Internal Point:** 0 tangents possible (lines are secants).
- **On-Circle Point:** Exactly 1 tangent possible.
- **External Point:** Exactly **2 tangents** can be drawn.

Length of a Tangent

- The length of the segment of the tangent from the external point to the point of contact is called the length of the tangent.

Section 10.3 Live Check

Q1. How many tangents can be drawn from a point lying inside a circle?

Q2. How many tangents can be drawn from a point on the circle itself?

Q3. How many tangents can be drawn to a circle from an external point?

Theorem 10.2 Statement & Proof

Theorem 10.2

The lengths of tangents drawn from an external point to a circle are **equal**.

Proof via Congruency

- Join external point P to centre O, and to points of contact Q and R (PQ, PR are tangents).
- In right triangles OQP and ORP: OQ = OR (radii), OP = OP (common), ∠OQP = ∠ORP = 90°.
- By RHS congruence, △OQP ≅ △ORP. Thus, **PQ = PR** (CPCT). Also ∠OPQ = ∠OPR (angle bisector).

Theorem 10.2 Live Check

Q1. State Theorem 10.2 regarding tangents drawn from an external point.

Q2. Which congruence criterion is used to prove that the two tangent segments are equal?

Q3. What additional geometric property does the line joining the external point to the centre possess regarding the angle between tangents?

Worked Exercise Highlights

Problem 1: Tangent Length (Ex 10.2 Q1)

- Tangent length PQ = 24 cm, distance OQ = 25 cm from centre.
- Radius OP ⊥ PQ, so $OP^2 = OQ^2 - PQ^2 = 25^2 - 24^2 = 657 - 576 = 49$.
- **Radius r = 7 cm.**

Problem 2: Quadrilateral Sums (Ex 10.2 Q8)

- Circumscribed quadrilateral ABCD. Tangents from external points are equal (AP=AS, BP=BQ, etc.).
- Adding opposite side segments: (AB + CD) = (AP+PB) + (DR+RC) = (AS+BQ) + (DS+CQ) = (AS+DS) + (BQ+CQ).
- **Proved: AB + CD = AD + BC.**

Exercise Highlights Live Check

Q1. How do we form a right-angled triangle when given the distance from the centre and tangent length?

Q2. In quadrilateral circumscribing circle proofs, which theorem allows us to equate tangent segments from vertices A, B, C, D?

Q3. Why are opposite angles formed by circumscribing quadrilaterals at the centre supplementary?

Chapter Recap & Checklist

Tangent Basics

- **Definition:** A line intersecting a circle at exactly one point.
- **Theorem 10.1:** Tangent is perpendicular to the radius through the point of contact.

External Tangents

- **Theorem 10.2:** Lengths of tangents drawn from an external point to a circle are equal.
- **Number of Tangents:** 0 from interior, 1 from on-circle, 2 from exterior point.

THANK YOU

"The circle is the first, the simplest, the most perfect figure." — Proclus

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